$+$ $\times$

$-$ $\div$

$2^5 \; 2^4 \; 2^3 \; 2^2 \; 2^1 \; 2^0$

INDIVIDUAL 1: 01 | 01 | 01 $\quad f(\text{IND.1}) = 2^0 + 2^2 + 2^4 = 21 \Rightarrow -21$

INDIVIDUAL 2: 10 | 10 | 10 $\quad f(\text{IND.2}) = 2^1 + 2^3 + 2^5 = 42 \Rightarrow -42$

INDIVIDUAL 3: 011001 $\quad f(\text{IND.3}) = 2^0 + 2^3 + 2^4 = 25 \Rightarrow -25$

INDIVIDUAL 4: 100110 $\quad f(\text{IND.4}) = 2^1 + 2^2 + 2^5 = 38 \Rightarrow -38$

- APPLY 2 POINT CROSSOVER
- APPLY SELECTION $f(x) = -x$

I CHOOSE IND.1 AND IND.3

DA FARE ACTRI

$f(x) = x_3 - 2x_2 + 6x_1 + x_0$

	$x_3 x_2 x_1 x_0$			
IND. 1	1010	$f(\text{IND.1}) = 1 - 0 + 6 + 0 = 7$	$P(\text{IND.1}) = 7/21 = 0.33$	TAKE IND.1 1 TIME
IND. 2	0111	$f(\text{IND.2}) = 0 - 2 + 6 + 0 = 5$	$P(\text{IND.2}) = 5/21 = 0.23$	TAKE IND.2 1 TIME
IND. 3	0011	$f(\text{IND.3}) = 0 - 0 + 6 + 1 = 7$	$P(\text{IND.3}) = 7/21 = 0.33$	TAKE IND.3 2 TIMES
IND. 4	1001	$f(\text{IND.4}) = 1 - 0 + 0 + 1 = 2$	$P(\text{IND.4}) = 2/21 = 0.09$	TAKE IND.4 0 TIMES

1

$f(\text{TOT}) = f(\text{IND.1}) + f(\text{IND.2}) + f(\text{IND.3}) + f(\text{IND.4}) = 7 + 5 + 7 + 2 = 21$

IND. 1 10|10 IND. 2 011|1

IND. 3 00|11 IND. 3 001|1

1 POINT CROSSOVER

$\Downarrow$

IND. 5 1011 IND. 7 0111

IND. 6 0010 IND. 8 0011

BIT MUTATION WITH $P = 1/4$

$\Downarrow$

IND. 5 1111 IND. 7 0111

IND. 6 1110 IND. 8 0010

AGE BASE SURVIVAL STRATEGY

NEW GENERATION $\Rightarrow$ IND. 5 1111 IND. 7 0111

IND. 6 1110 IND. 8 0010